Did COVID-19 Increase Corruption Risk in Public Procurement?

Evidence from Mexico's Federal Procurement System

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Public Summary

Governments worldwide responded to COVID-19 by **rapidly increasing procurement budgets** while simultaneously **relaxing accountability standards** to speed up purchasing.

The natural expectation: **healthcare institutions** would be the corruption hotspot, most money, most relaxed rules, most pressure.

The research question: Did the pandemic measurably increase **corruption risk** in Mexico's federal procurement, and if so, **where?**

The answer: The increase was driven almost entirely by **non-healthcare institutions**, not by healthcare.

The outcome variable: DCVB

Mexico allocates public contracts through three mechanisms: open auctions, invitation auctions, and direct allocation.

The **DCVB ratio** measures the share of total contract value assigned through discretionary mechanisms (invitation or direct): $\text{DCVB}_{im} = \frac{\text{Discr. value}_{im}}{\text{Total value}_{im}}$

A higher DCVB signals **greater corruption risk**.

Data from **CompraNet**, Mexico's public procurement system, systematized by the Instituto Mexicano para la Competitividad (IMCO).

Feature	Value
Federal institutions	64
Public acquisitions	378,000
Institution-month obs.	2,304
Period	Jan 2018 – Dec 2020
DCVB (Overall)	0.705
DCVB (Health)	0.667
DCVB (Non-Health)	0.715

Source: IMCO. Pre-pandemic means.

Two groups of institutions:

Healthcare: federal hospitals, IMSS, and health secretariats. Under intense public scrutiny during the pandemic.

Non-healthcare: defense, education, infrastructure, energy, and other sectors. Comparatively less media attention.

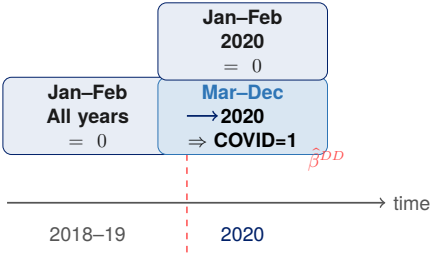
Before the pandemic, **70% of contract value** was already allocated through discretionary mechanisms. This is the baseline the pandemic disrupted.

The pandemic is a **universal shock**: all 64 institutions were affected simultaneously. Variation comes from the **time dimension**:

- **Treatment indicator:** 1 if year = 2020, 0 for 2018–2019
- **Time indicator:** 1 for March–December, 0 for January–February
- **COVID-19 variable:** interaction of the two above

$$\text{DCVB}_{imy} = \alpha + \beta^{DD}\,\text{COVID19}_{imy} + a_i + \gamma_m + \nu_y + e_{imy}$$

FE at institution (a_i), month (γ_m), year (ν_y). SE clustered at institution level.



An **event study** traces the effect month by month and validates the parallel trends assumption.

Difference-in-Differences results (Table 2):

	Overall	Health	Non-Health
1(COVID-19)	**0.120*** **	−0.060	**0.166*** **
	(0.034)	(0.043)	(0.039)
N	2,304	468	1,836
R^2	0.32	0.19	0.35
Pre-COVID mean	0.705	0.667	0.715
% change	**+17.02%**	−9.09%	**+23.21%**

*** $p < 0.01$. Baseline FE at institution, month, and year level. Source: IMCO.

The counterintuitive result:

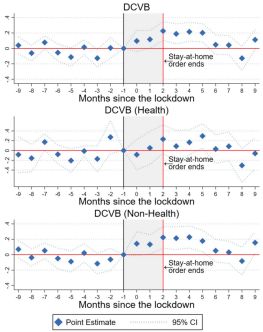
The pandemic increased discretionary contracting by **+17%** overall.

This is driven entirely by **non-healthcare institutions (+23%).**

Healthcare institutions show **no significant change**, despite being at the center of the public debate.

A possible explanation: intense media scrutiny of healthcare procurement may have **constrained discretionary behavior** precisely where it was most expected.

Event study (Figure 1):



Pre-pandemic coefficients near zero (parallel trends). Non-health DCVB peaks at $q = 2$ (May 2020), fades within 6 months. Healthcare DCVB flat throughout.

Two mechanisms investigated:

Market concentration (HHI):

Each institution's supplier market is measured with a Herfindahl-Hirschman Index. Non-healthcare institutions saw their supplier market become significantly more concentrated.

Late disclosure of contract information:

The proportion of contracts with procedural information published after the contract start date increased for non-healthcare institutions, consistent with reduced transparency.

Leave-one-out analysis and exclusion of SEDENA confirm robustness.

What we found:

- COVID-19 increased discretionary contracting in Mexico's federal procurement by **+17%** on average
- The effect is driven entirely by **non-healthcare institutions (+23%),** not by healthcare
- The increase was **temporary:** peaked two months after the stay-at-home order was lifted, returned to baseline within six months
- Non-healthcare institutions simultaneously became more **concentrated** in their supplier base and less **transparent** in contract disclosure

Limitation: DCVB measures *corruption risk*, not actual corruption. The paper identifies a pattern at the aggregate level, not individual wrongdoing.

Policy implication:

Audit efforts targeting pandemic-era procurement should focus on **non-healthcare institutions,** even though the political debate centered on healthcare.

Public scrutiny may itself be a **corruption deterrent:** sectors that receive less attention are where the risk actually materializes.

Silverio-Murillo, Prudencio & Balmori-de-la-Miyar (2024), *Public Organization Review* 24, 1293–1313.